
The Empirics of Agglomeration

LSE GY457 - AT 2025

Topic 2

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- **Topic 2: The empirics of agglomeration**
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- Block 3

- Topic 7: Neighborhoods and segregation
- Topic 8: The vertical dimension of cities
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Incentives for spatial clustering in regions
 - Increasing returns + low transportation costs (for goods)
- Incentives for spatial clustering in cities
 - External returns + high transportation costs (for people)
- Impact on real estate
 - Clustering increases the demand for real estate, shifting the demand curve
 - Changes in productivity origins/sources affect future demand

Today: The empirics of agglomeration

- The empirics of agglomeration
 - Measurement
 - Clustering and the determinants of agglomeration
 - Spatial scale of agglomeration effects
- The effects of agglomeration
- Urbanization vs. localization effects
 - What matters for urban growth

Today: The empirics of agglomeration

- The empirics of agglomeration
 - Measurement: *How do we measure (spatial) concentration?*
 - Clustering and the determinants of agglomeration
 - Spatial scale of agglomeration effects
- The effects of agglomeration
- Urbanization vs. localization effects
 - What matters for urban growth

Measures of (spatial) concentration

- What do we need to build a measure of concentration?
And of *spatial* concentration?
 - Distribution of employment shares across space (x_i), e.g. $\sum x_i^2$
How uneven is the country's employment?
 - Herfindahl-Hirschman Index $H \equiv \sum_i s_i^2$
How is the industry's (establishment) concentration?
 - (Spatial) Gini coefficient $G \equiv \sum_i (x_i - s_i)^2$
How different is the industry's regional distribution from economy's?
 - Ellison-Glaeser (1997) Index (next slide)
- All these measures rely on choosing market/geographic scope, i.e. determining “How close is close” (more later)

Ellison-Glaeser (1997) Index

$$\gamma = \frac{G - (1 - \sum_i x_i^2)H}{(1 - \sum_i x_i^2)(1 - H)}$$

- Captures the extent to which one industry is more concentrated in space than overall employment
 - $\gamma = 0$ when industry is concentrated as in case of random location process
 - $\gamma > 0$ captures the degree of relative spatial concentration
 - BUT: does not mean agglomeration externalities are necessarily present (cannot identify the *causes* of concentration)

Practical example

region	workers industry A	workers industry B	totals
N	4	2	6
W	16	2	18
S	0	3	3
E	0	3	3
totals	20	10	30
$\sum x_i^2$	?		
$\sum s_i^2$ (A)	?		
$\sum s_i^2$ (B)	?		
$G = \sum_i (x_i - s_i)^2$ (A)	?		
$G = \sum_i (x_i - s_i)^2$ (B)	?		
EG index (A)	?		
EG index (B)	?		

To solve and try different scenarios click [here](#)

Today: The empirics of agglomeration

- The empirics of agglomeration
 - Measurement
 - Clustering and the determinants of agglomeration: *Which industries are more likely to concentrate in space and why?*
 - Spatial scale of agglomeration effects
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Which industries agglomerate? (Rosenthal and Strange, 2001)

Agglomeration of Manufacturing Industries at the SIC 2-Digit Level

SIC	Definition	State γ	County γ	Zip code γ
20	Food and kindred products	0.00347	0.00119	0.00029
21	Tobacco manufactures	0.19457	0.13002	0.03989
22	Textile mill products	0.09410	0.00601	0.00177
23	Apparel and related products	0.01159	0.00653	0.00184
24	Lumber and wood products, except furniture	0.01168	0.00284	0.00034
25	Furniture and fixtures	0.01212	0.00297	0.00074
26	Paper and allied products	0.00844	0.00213	0.00035
27	Printing, publishing, and allied products	0.00527	0.00264	0.00039
28	Chemicals and allied products	0.01047	0.00369	0.00062
29	Petroleum refining and related products	0.03605	0.01040	0.00428
30	Rubber and miscellaneous plastics products	0.00385	0.00102	0.00023
31	Leather and leather products	0.01513	0.00640	0.00298
32	Stone, clay, glass, and concrete products	0.00357	0.00209	0.00052
33	Primary metal products	0.01438	0.00202	0.00041
34	Fabricated metal products, except machinery and transportation equipment	0.00447	0.00095	0.00021
35	Machinery, except electrical	0.00170	0.00112	0.00029
36	Electrical and electronic machinery, equipment, and supplies	0.00869	0.00352	0.00050
37	Transportation equipment	0.02203	0.00462	0.00084
38	Scientific and professional instruments; photographic and optical goods; watches	0.01453	0.00429	0.00018
39	Miscellaneous manufactured commodities	0.00666	0.00306	0.00055

- All industries have positive Ellison-Glaeser Index (concentrated)
- Highest level of concentration at the state level

The 10 most concentrated industries

The 10 Most Agglomerated Manufacturing Industries at the SIC 4-Digit Level

Zipcode level			County level			State level		
SIC	SIC description	γ	SIC	SIC description	γ	SIC	SIC description	γ
2371	Fur goods	0.352	2371	Fur goods	0.372	2397	Schiffli machine embroideries	0.499
3761	Guided missiles & space vehicles, parts	0.260	2397	Schiffli machine embroideries	0.251	3761	Guided missiles & space vehicles, parts	0.434
3579	Office machines & parts	0.145	3761	Guided missiles & space vehicles, parts	0.239	2284	Thread and handwork yarns	0.413
2087	Flavoring extracts & syrups	0.142	2874	Phosphatic fertilizers	0.215	2371	Fur goods	0.408
3149	Footwear, except rubber, n.e.c.	0.139	3861	Photographic equipment and supplies	0.191	2273	Carpets and rugs	0.406
2335	Womens' and misses' dresses	0.118	2111	Cigarettes	0.191	2084	Wines, brandy, and brandy spirits	0.372
2381	Fabric dress and work gloves	0.114	3149	Footwear, except rubber, n.e.c.	0.185	2251	Womens' hosiery, except socks	0.371
3764	Missile and rocket engines	0.111	2043	Cereal breakfast foods	0.180	3533	Oil & gas field equipment & parts	0.339
3676	Electronic resistors	0.086	2335	Women's, misses', and juniors'	0.178	2436	Softwood veneer and plywood	0.328
3844	X-ray apparatus, tubes, & parts	0.084	2841	Soap & detergents	0.167	2141	Manufactured tobacco	0.305

Scope for location fundamentals?
Agglomeration externalities?

The determinants of agglomeration

Three key sources of agglomeration spillovers (Rosenthal and Strange, 2001):

1. Knowledge spillovers
2. Labor market pooling
3. Input sharing

Plus:

4. Location fundamentals (natural advantages)
5. Transportation costs

Proxies for natural advantages and transportation costs

- Natural advantages impact location decisions of firms because affect cost of shipping inputs (and outputs)
 - In turn, affect concentration as firms locate close to where the sources of their inputs are
 - Industries tied to locations with cheap energy/resources (e.g., measured by *energy expenses per \$ shipment*)
 - Cost of transporting output matters too
 - Industries with output that is costly to transport are more likely to locate closer to their markets (may lead to less agglomeration)
 - Proxied by *(end-of-year) inventories per \$ shipment*: low inventories >> perishable/bulky products >> locate near buyers
-

Proxies for agglomeration externalities

- Input sharing (suppliers/services)
 - *Manufactured inputs per \$ shipment* ↑ relies on specialized parts >>> benefits from cluster
 - *Non-manufactured (services) inputs per \$ shipment* ↑ (like legal services) >>> weaker benefit to cluster
- Knowledge spillovers: *Innovations (new products) per \$ shipment*

Proxies for agglomeration externalities

- Labor market pooling: *Net productivity per worker* $\uparrow$, *share of management (over management + production workers)* $\uparrow$, and *share of high-skilled workers* $\uparrow$
 - If labor pooling possible, an industry benefits from agglomeration because is better able to hire workers with industry-specific skills

How to identify the origins of agglomeration?

- Regress these industry-level proxies on spatial concentration indexes:

$$\gamma_{j,m} = \beta X_m + \varepsilon_{j,m}$$

- j spatial level (state, county, zip code)
- m industry
- X vector of variables to proxy determinants of agglomeration
- β vector of parameters capturing marginal effect of X on γ
- Unit of analysis: industry

Main takeaways (Rosenthal and Strange, 2001)

Determinant/Geography	Zip code	County	State
Location fundamentals	X	X	√
Transportation cost	X	X	√
Input sharing	X	X	√
Labor market pooling	√	√	√
Knowledge spillovers	√	X	X

√ = Suggestive evidence; X = No/weak evidence

- Location fundamentals, transportation costs, and input sharing matter at *regional* level
- Labor market pooling matters at *all levels*
- Knowledge spillovers matter at *local* level

Today: The empirics of agglomeration

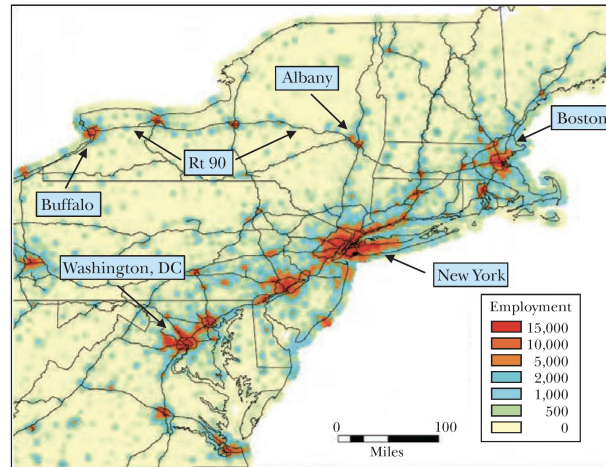
- The empirics of agglomeration
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Spatial scale of agglomeration effects

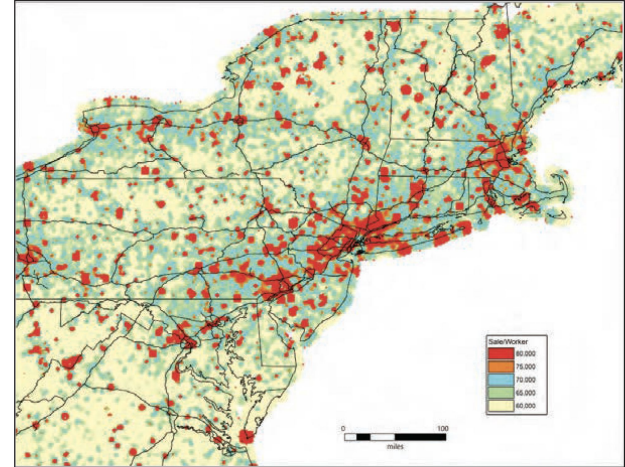
- How close must workers and firms be to benefit from agglomeration economies?
- “How close is close?” (Rosenthal and Strange, 2020)
 - Implicit in the idea that spatial concentration increases productivity is that the *degree of proximity* matters
 - Agglomeration effects operate even below the neighborhood level, within buildings and organizations
- Important also for policy: How to internalize benefits
- Crucial for markets like commercial real estate and transportation
 - Agglomeration economies capitalized in rents and prices, and affect the design of transportation networks

“How close is close?” (Rosenthal and Strange, 2020)

A: Aggregate employment



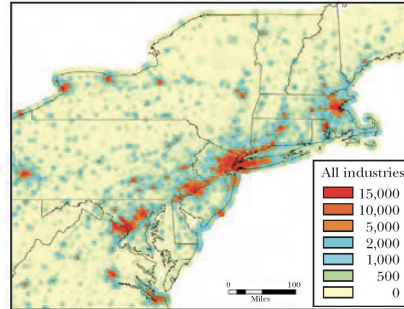
B: Single-site sale per worker



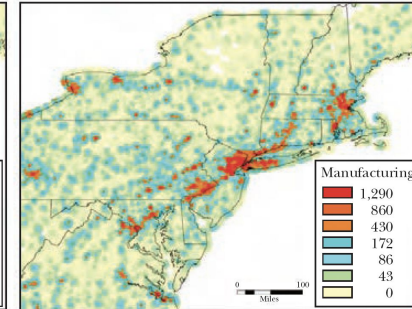
Data source: Dun & Bradstreet establishment data (retrieved in June 2019)

“How close is close?” (Rosenthal and Strange, 2020)

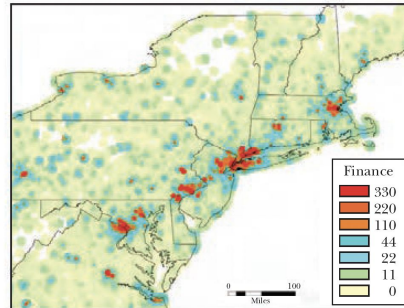
A: All industries



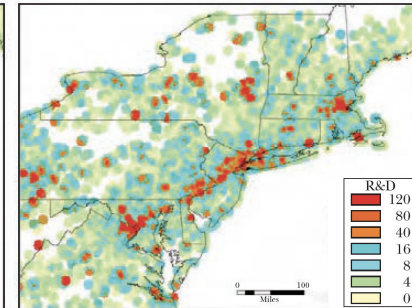
B: Manufacturing



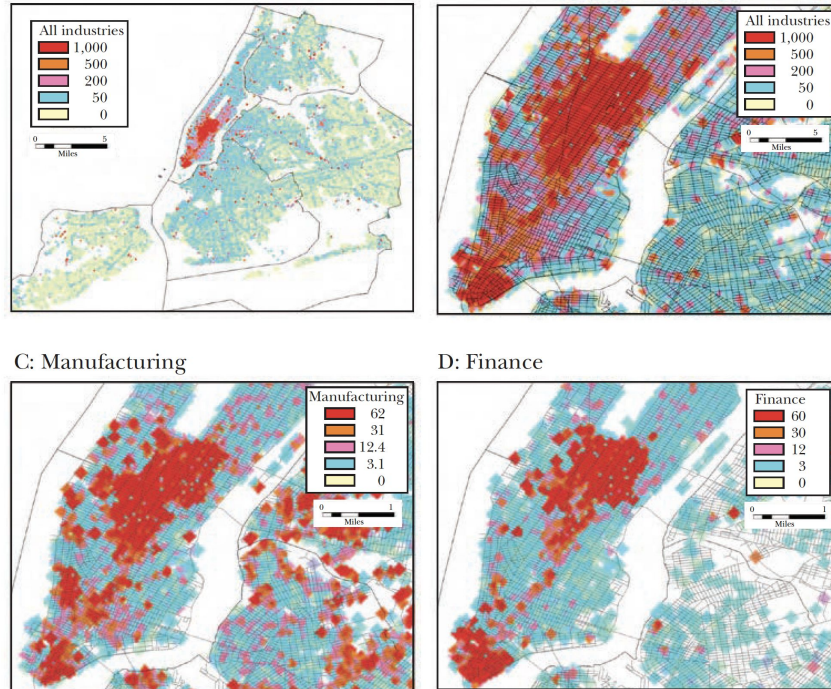
C: Finance



D: R&D



“How close is close?” (Rosenthal and Strange, 2020)



“How close is close?” (Rosenthal and Strange, 2020)

- Agglomeration effects operate at various spatial scales, with nearby effects being the strongest
- Mechanisms explaining this pattern:
 - Single agglomeration effect (spillovers decreasing with distance)
 - Different agglomeration forces operate at different spatial levels (e.g., interplay of knowledge spillovers and labor market pooling effects)
 - Heterogeneous interaction costs (proximity is beneficial to all agents, but the ability to leverage distant interactions varies with networking capabilities)

“How close is close?” (Rosenthal and Strange, 2020)

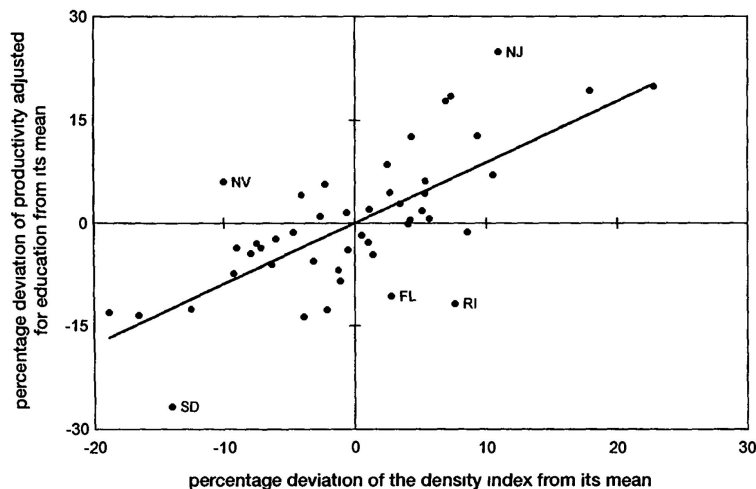
- Labor market pooling matters within *commuting areas*
 - Metropolitan areas defined in part by commuting flows
- Knowledge spillovers operate at narrower levels
 - Large degree of spatial decay, at neighborhood or lower levels
 - e.g., Rosenthal and Strange (2008), Arzaghi and Henderson (2008), Liu et al. (2018; 2024), Mas and Moretti (2009)

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The effects of agglomeration

- Do increases in density increase productivity? (Ciccone and Hall, 1996)
- Density elasticity of productivity is 0.06



- Cities with higher density are more productive
- Doubling density increases productivity by 4.2% ($= 2^{0.06} - 1$)
- *Any challenges to identification?*

Individual-level vs city-level estimates

- Individual-level wage equation:

$$W_{i,c} = u_i + \gamma Z_c + \theta X_i + \varepsilon_{i,c}$$

- u_i is individual-level unobserved fixed effect
- Z_c is local density
- X_i is observed individual-level attribute (e.g., education)
- From Combes and Gobillon (2015)

- City-level wage equation:

$$\bar{W}_c = \gamma Z_c + \theta \bar{X}_c + \varepsilon_c$$

- where $\varepsilon_c = \bar{\eta}_c + \bar{u}_c$ and upper bars denote city averages

City-level estimates

- Many studies estimate $\bar{W}_c = \gamma Z_c + \theta \bar{X}_c + \varepsilon_c$ ($\varepsilon_c = \bar{\eta}_c + \bar{u}_c$)
- But: unobserved individual-level effects are a concern

$$\text{Cov}(\bar{u}_c, Z_c) \neq 0$$

- People with unobserved abilities sort into large and dense cities to benefit from job opportunities
- Empirically, γ between 0.04 and 0.07
- Interpret them as correlations, not as causal effects of density on productivity (selection matters!)

Individual-level estimates

- With individual data, we can estimate $W_{i,c} = u_i + \gamma Z_c + \theta X_i + \varepsilon_{i,c}$
- Adding individual-level fixed effects, we control for unobserved individual characteristics
- Identification comes from “movers” i.e. from comparing the wages of the same individual in different areas
- Controlling for fixed effects, the elasticity of productivity drops to around $\gamma = 0.02$
- Closer to causal interpretation, since controls for sorting
- For policy: what is the “pure” effect of density

City-level: IV regression

- Another concern is unobserved location effects

$$\bar{W}_c = \gamma Z_c + \theta \bar{X}_c + \varepsilon_c \quad \varepsilon_c = \bar{\eta}_c + \bar{u}_c$$

- If $Cov(\bar{\eta}_c, Z_c) \neq 0$, individual level fixed effects will not solve the problem
- Solution: IV regression (two-stage least squares)
 - First stage: $Z_c = \varphi V_c + \theta X_i + u_i + v_{i,c}$
 - Use predicted values from first stage in the second stage
 - Second stage: $W_{i,c} = u_i + \gamma \hat{Z}_c + \theta X_i + \varepsilon_{i,c}$
 - Requires $Cov(V_c, Z_c) \neq 0, Cov(V_c, \eta_c) = 0$

Side notes: Good IVs

- Good Instrumental Variables need to satisfy:
 - Relevance: $Cov(V_c, Z_c) \neq 0$
 - Exclusion restriction: $Cov(V_c, \eta_c) = 0$
- Difficult to find! Even more difficult, need a good argument (no statistical test can help)
- For instance, Ciccone and Hall (1996) use *historical measures of density*:
 - Presence of railroad in 1860
 - State population in 1850 and 1880
 - Distance from eastern seaboard

Unobserved factors with panel data

- Use time dimension of the data to remove the unobserved factors (*very* common strategy)

$$W_{i,c,t} = u_i + \gamma^{\text{MP}} \tilde{Z}_{c,t} + \theta X_{i,t} + \rho_t + \varepsilon_{i,c,t}$$

- e.g., Donaldson-Hornbeck (2016), Ahlfeldt-Feddersen (2018)
- $\tilde{Z}$ is measure of “economic mass” (GDP, employment) within a certain travel time (or a market potential measure)
- If transport infrastructure changes, we can take time differences (Δ):

$$\Delta W_{i,c} = \gamma^{\text{MP}} \Delta \tilde{Z}_c + \theta \Delta X_i + \rho + \Delta \varepsilon_{i,c}$$

- Unobserved (time-invariant) factors are removed!
- Ahlfeldt and Feddersen (2018) find $\gamma = 2\text{-}3\%$

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Localization economies

- About spillovers between firms *in an industry*
- MAR: Marshall (1890) – Arrow (1962) – Romer (1986)
 - Focus: Specialization
 - Examples: Autos in Detroit, film industry in LA
 - Implications: Government should welcome local monopolies since they lead to innovation (internalize externalities)
- Porter (1990)
 - Focus: Specialization and competition
 - Examples: Italian ceramics and gold industry
 - Implications: Governments should encourage geographically localized competitive industries

Urbanization economies

- About spillovers from outside the industry
- Jacobs (1969)
 - Focus: Diversity
 - Examples: Gas engine firms in Detroit supplied shipbuilders and later car makers
 - Implications: Governments should encourage a variety and diversity of geographically proximate industries

Which theory(-ies) is empirically most relevant?

Empirical findings: Henderson (1988)

- Multivariate regression of the form:

$$q_{i,j} = \beta_0 + \beta_1 Q_{i,j} + \beta_2 N_j + X_{i,j} \theta + \varepsilon_{i,j}$$

- $q_{i,j}$ is output per worker in industry i and city j
 - $Q_{i,j}$ total industry i output in city j (proxy for localization)
 - N_j total population/employment in city j (proxy for urbanization)
 - $X_{i,j}$ vector of industry-city controls (education, capital per worker)
- $\hat{\beta}_1$ depends strongly on industry
 - Min = 0.02 (pulp and paper); Max = 0.11 (petroleum)
- $\hat{\beta}_2$ close to zero
- Localization (not urbanization) good for productivity

Empirical findings: Glaeser et al. (1992)

- Regress 1956-1987 city by-industry employment growth against measures of specialization and competition
 - Specialization = $\frac{\text{industry empl city}}{\text{total empl city}} / \frac{\text{industry empl US}}{\text{total empl US}}$
 - Competition = $\frac{\text{firms in city} - \text{industry}}{\text{workers in city} - \text{industry}} / \frac{\text{firms in US industry}}{\text{workers in US industry}}$
- Competition good for economic growth
- Specialization negative effects on growth
- Small industries grow if large ones grow
- Supports Porter & Jacobs, not MAR

Empirical findings: Henderson et al. (1995)

- Regress 1987 (log) city by-industry employment on
 - (log) 1970 employment (persistence vs. convergence)
 - 1970 industry/city employment (specialization)
 - 1970 HHI (inversely related to urbanization)
- Separate analyses by type of industries:
 - Mature: (electrical) machinery, primary metals, transportation
 - High-tech: Electronic components, medical equipment, computers
- Mature industries support MAR
 - Localization, not urbanization economies promote growth
- High-tech industries support MAR & Jacobs
 - Localization and urbanization economies promote growth

Empirical findings: Duranton and Puga (2001)

- Nursery cities
 - New products developed in diversified cities
 - Try prototypes using processes borrowed from different activities
 - Specialized cities
 - Once ideal production process found, firms/plants relocate
 - Mass production in specialized cities with lower production costs
 - Evidence supports theory:
 - 72% of relocations happen from diversified to specialized areas
 - 93% in R&D, 88% in pharma, 82% in IT sectors (France, 1993-1996)
 - Mature industries support localization (MAR)
 - High-tech industries support urbanization (Jacobs)
-

Takeaways of empirical findings

- Both urbanization and localization economies matter empirically
- Localization relevant for productivity levels
- Urbanization important for growth, especially of innovative high-tech industries
- Rationalize empirical fact that often small specialized cities are quite productive (high wages), but large and diversified cities tend to be more dynamic

Summary

- Empirics of agglomeration
 - How to measure spatial concentration
 - Which industries concentrate and why
 - How large are spatial spillovers
- Estimates of effects of agglomeration on productivity
 - Doubling density increases area productivity by ~4% on average
 - Doubling density increases individual productivity by ~2%
- Urbanization vs. localization effects
 - Localization economies impact distribution of productivity levels
 - Urbanization economies matter for growth

Next week

- The economic effects of density
 - How to measure cities
 - Costs and benefits of density
- Equilibrium city size
 - Optimal city size
 - Inefficiencies and policies
- Compensating differentials
 - Rosen-Roback framework

Thank you!

Readings and references

Required readings

- Combes, P. P., and Gobillon, L. (2015). The Empirics of Agglomeration Economies. In G. Duranton, J. V. Henderson, and W. C. Strange (eds). Handbook of Regional and Urban Economics, Vol. 5, 247–348. Amsterdam: Elsevier.
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